

Basic Principles of Medicine 1
Module: Cardiovascular, Pulmonary, Renal
DLA 14

Lung Function & Respiratory Gases 1

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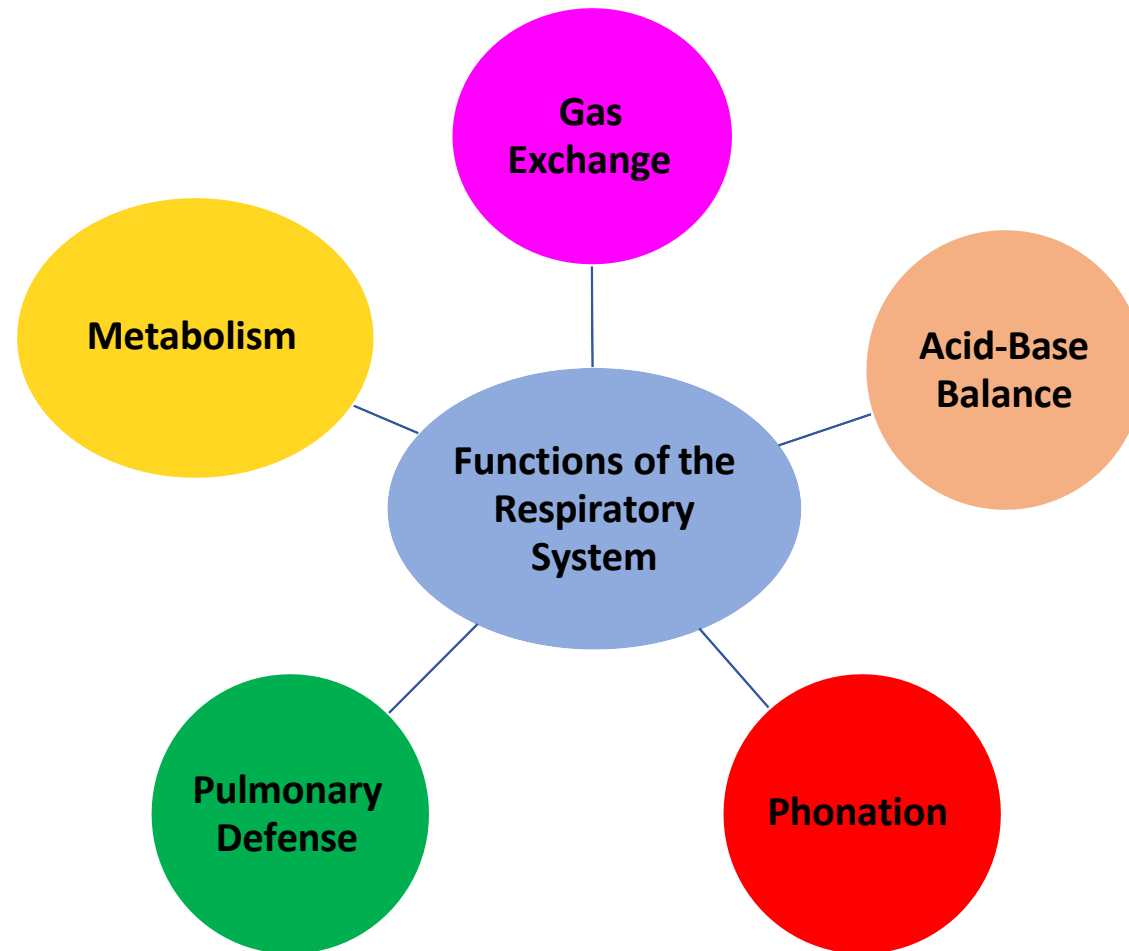
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Learning Objectives

By the end pf this DLA, the student should be able to

SOM.MK.II.BPM1.3.CPR.2.PHYS.0191	Describe the structural characteristics of airways.
SOM.MK.II.BPM1.3.CPR.2.PHYS.0192	Describe the transport of gas through the conducting airways to and from the alveoli.
SOM.MK.II.BPM1.3.CPR.2.PHYS.0193	Explain the key roles of mucociliary transport and describe related pathologies and their clinical presentation.
SOM.MK.II.BPM1.3.CPR.2.PHYS.0194	Describe the clinical significance of mucociliary transport.
SOM.MK.II.BPM1.3.CPR.2.PHYS.0195	Describe the functions of alveolar Type 1 & type II cells.
SOM.MK.II.BPM1.3.CPR.2.PHYS.0196	Lists components of chest wall and lungs and relate the functions of respiratory muscles to movement of air in and out of the alveoli.
SOM.MK.II.BPM1.3.CPR.2.PHYS.0197	Describe how Boyle's and Charle's law apply to movement of gas in and out of lungs.
SOM.MK.II.BPM1.3.CPR.2.PHYS.0198	List the metabolic functions of the lungs and explain their key physiological roles.

Respiratory System – General Functions



Metabolic Functions

Metabolism of vasoactive hormones in the lung are selective

Activation	Inactivation
Angiotensin I  Angiotensin II	- Bradykinin - Serotonin - Prostaglandin E1,E2,& F2

Conduction and Respiratory Zone

Respiratory zone

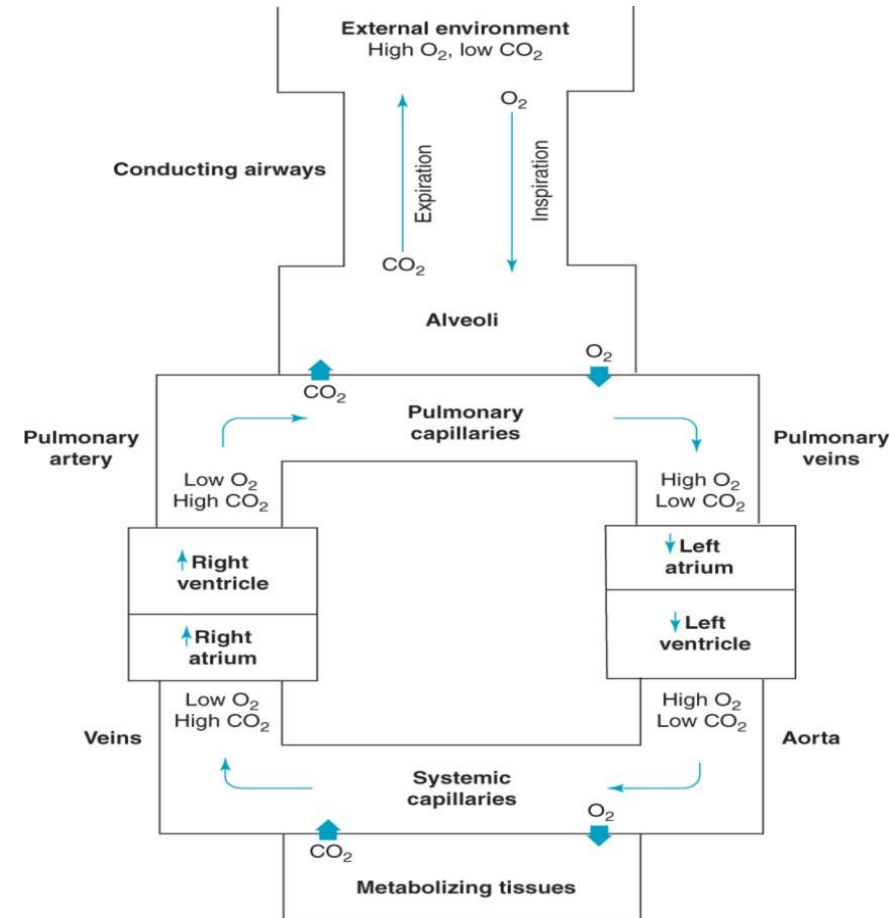
- End of the respiratory tree (respiratory bronchioles, alveolar ducts, and alveolar sacs)
- Gas-exchange surface
- Epithelial layer and an extracellular matrix
- Surrounded by capillaries

Conducting zone

- Air flow into and out of the lungs (nose to terminal bronchioles)
- Heat, hydrate and clean
- Heat exchange, water vapor pressure equilibration, and remove particulate load.
- Mucociliary clearance

Airway and Gas Transport

- Gas moves from the trachea to the alveoli via **bulk flow**
- In the alveoli gas moves across the alveolar-capillary membrane by **diffusion**



Source: Levitzky MG: Pulmonary Physiology, Eighth Edition:
www.accessmedicine.com
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Source: Pulmonary Physiology, *Medical Physiology: The Big Picture*

Pulmonary Blood-Gas Barrier

Fick's Law

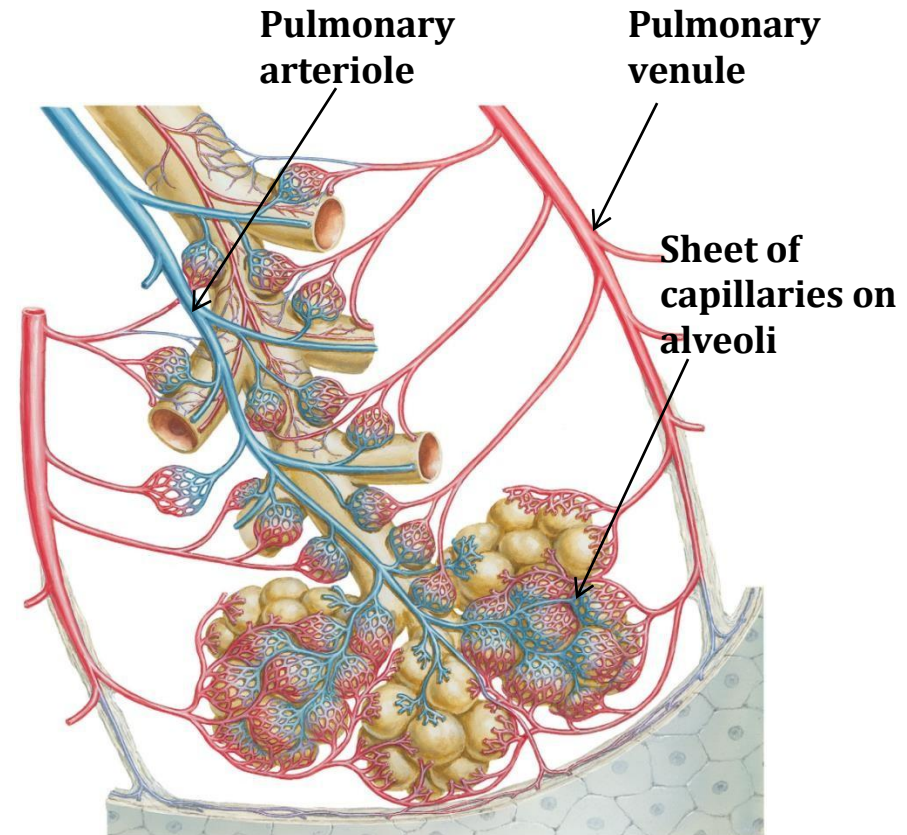
$$\dot{V}_{\text{gas}} = D \frac{A}{t} (P_1 - P_2)$$

- V – rate of gas diffusion
- A – surface area available for exchange of gas
- T- diffusion barrier thickness
- D – diffusion coefficient

Respiratory System – Pulmonary Blood-Gas Barrier

Surface Area

- Gas exchange occurs in the over 300 million closed air sacs called **alveoli** (75m²)
- Pulmonary circulation – most extensive network of capillaries in the body



Pulmonary Blood-Gas Barrier

Diffusion Barrier Thickness

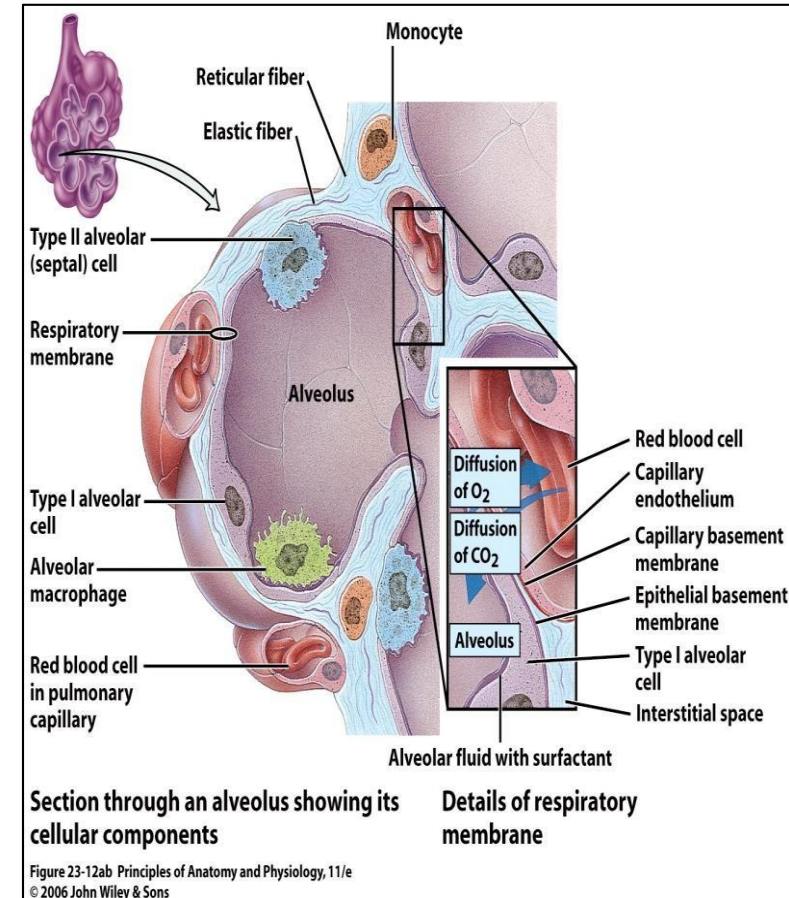
- The pulmonary blood-gas interface is about 0.5- μm thick
- Formed by the alveolar-capillary membrane of the terminal respiratory unit.

Consists of:

1. Thin layer of surface liquid.
2. Alveolar epithelium cells (type 1 pneumocytes), plus associated basement membrane.
3. Thin layer of interstitial fluid.
4. Pulmonary capillary endothelial cells, plus associated basement membrane.

Alveolar cells

- Two types of cells in alveolar epithelium
 - Type 1 pneumocytes
 - Type 2 pneumocytes
- Ratio is 1:1
- Type 1 pneumocytes occupy most of the surface area
- Type 2 pneumocytes make surfactant



Alveolar cells & Surfactant

- Surface Active Agent
- Secreted by type II pneumocytes
- Main component
- Dipalmitoylphosphatidylcholine
- **Reduces surface tension greatly – increasing lung compliance**

